

Surya Partap Singh

Senior Data Engineer · Azure · Databricks · Spark · Lakehouse Platforms

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SUMMARY

Senior Data Engineer with 5+ years building and scaling enterprise lakehouse platforms on **Databricks, Delta Lake, Apache Spark, and Azure** — unifying clinical, financial, and pharmacy data for analytics, reporting, and ML. Proven track record of optimizing distributed workloads, automating CI/CD pipelines, and mentoring and leveling up engineering teams in high-stakes, regulated production environments.

TECHNICAL SKILLS

Programming: Python, PySpark, Spark SQL, SQL, C++

Data Platforms: Databricks, Delta Lake, Apache Spark, Hadoop, Hive, HDFS

Cloud & Storage: Microsoft Azure (Data Lake, Blob Storage, Azure SQL, ADF), Amazon S3

Streaming: Spark Structured Streaming, Delta Change Data Feed

Orchestration: Databricks Workflows, Azure Data Factory, Airflow

Data Quality & Governance: Great Expectations, schema evolution, anomaly detection, data lineage, observability

DevOps & IaC: GitHub Actions, Azure DevOps, ARM Templates, Git, Docker

Ways of Working: Agile, code reviews, mentoring, production support

EXPERIENCE

Cencora

Senior Data Engineer, Data Platform

Dallas, TX

Feb 2023 – Present

- **Lakehouse Platform Architecture:** Designed and scaled a **Databricks** and **Delta Lake**-based medallion lakehouse for clinical, financial, and pharmacy data; built reusable ingestion and transformation frameworks with canonical data models, reducing time-to-insight by **50%** and onboarding time for new data sources by **40%**.
- **Spark Performance Optimization:** Improved end-to-end pipeline throughput by **30%** by tuning **Apache Spark** workloads across partitioning, shuffle behavior, join strategies, and cluster resource utilization.
- **Near-Real-Time Processing:** Reduced financial reporting latency from **24 hours to 15 minutes** by redesigning batch pipelines into incremental and near-real-time processing using **Delta CDF** and **Spark Structured Streaming**.
- **Unstructured Data Pipelines:** Built OCR-driven ingestion pipelines to extract and standardize data from unstructured patient documents, enabling curated datasets for clinical analytics and downstream ML workflows.
- **Data Quality & Observability:** Implemented automated data quality controls using **Great Expectations** and anomaly detection frameworks catching schema drift, freshness failures, and integrity issues before downstream impact reducing data incidents by **35%**.
- **Platform Automation:** Established CI/CD and infrastructure automation for data pipelines and cloud resources using **GitHub Actions**, **Azure DevOps**, and **ARM Templates**, improving release consistency and cutting deployment time by **25%**.
- **Cloud Cost Optimization:** Reduced Azure Databricks compute costs by **30%** through right-sizing cluster configurations, enforcing auto-termination policies, migrating low-priority workloads to spot instances, and consolidating redundant jobs delivering measurable cost reduction without impacting pipeline SLAs.
- **Unity Catalog & Data Governance:** Led adoption of **Databricks Unity Catalog** to enforce fine-grained access control, column-level security, and data lineage tracking across the lakehouse establishing a centralized governance layer that ensured regulatory compliance and audit-readiness across clinical and financial datasets.
- **Technical Leadership:** Mentored 3–5 engineers on ETL design, Spark optimization, and platform best practices defining team-wide engineering standards through structured code reviews and reusable documentation.
- **Engineer Onboarding:** Reduced new engineer ramp-up time by **30%** by building onboarding frameworks, internal wikis, and leading pair programming sessions.

Cencora (Contract)

Data Engineer

Dallas, TX

May 2021 – Feb 2023

- **Production Data Pipelines:** Built and supported production ETL pipelines on **Databricks**, **Azure Data Factory**, and **PySpark** for large-scale clinical, pharmacy, and financial datasets.
- **Canonical Data Models:** Designed canonical, single-source-of-truth tables that standardized disparate source systems improving cross-team reporting consistency and eliminating **3+** redundant dataset copies maintained across business units.

- **Distributed Data Processing:** Optimized large-scale **Spark** transformations using broadcast joins, predicate pushdown, and partition-aware reads to improve runtime efficiency and reduce compute costs.
- **Workflow Reliability:** Automated and monitored ETL workflows through **Databricks Jobs**, implementing retries, alerting, and operational checks improving pipeline success rates to **99%+** and ensuring consistent SLA adherence.

Fuge Technologies Inc

Data Engineer

Dallas-Fort Worth, TX

Feb 2021 – May 2021

- **Batch Processing:** Built and optimized Hadoop and Hive-based batch pipelines for large structured datasets, applying partition pruning, bucketing strategies, and execution plan analysis to improve query performance in distributed environments.
- **Azure Data Ingestion:** Developed ingestion pipelines into Azure Data Lake using **Azure Data Factory**, supporting incremental loads, schema evolution handling, and downstream validation checks for analytics use cases.
- **Data Standardization:** Collaborated with cross-functional teams to normalize and standardize raw source data into consistent, analytics-ready formats — establishing naming conventions and structural patterns adopted across the pipeline.

EDUCATION

University of North Alabama

Bachelor of Science in Computer Science

Florence, AL

Aug 2016 – Dec 2020

- **Relevant Coursework:** Distributed Systems, Database Management Systems, Data Structures & Algorithms, Operating Systems, Cloud Computing, Software Engineering, Machine Learning